



MATH 310 Probability and Statistics (L4 & L3)
Dr. Aamir Hasan & Ms. Ummamah Shah
Mid-Term Exam – Spring 2026
Exam Date: 03 March 2026

Total Marks: 60

Time: 90 min

Name: _____

ID: _____

Instructor: _____

Section: _____

Time allotted: 90 minutes

For examiner's use only

Question	Points	Score	Targeted CLOs
1	10		CLO-1
2	15		CLO-3
3	15		CLO-2
4	20		CLO-4
Total	60		

Instructions:

- Mobile phones and other electronic devices must be **turned off** and put away **before** the start of the exam. You are allowed to use a scientific calculator (**Not** the one on your smart phone. Sharing of calculators is **not** allowed as well).
- **No** food or drink, other than water, is allowed in the examination room.
- You are **not** allowed to communicate with other candidates.
- You are **not** allowed to leave your seat unless granted permission by an invigilator.
- All answers **must** be supported by proper working and justification.

Question 1 [10 points]

Probability and Independence

We roll a fair tetrahedral die (4-sided die with numbers 1, 2, 3 and 4) **three times**.

- (a) Calculate the probability that, after three rolls, the number 4 appears at least one time. 2 Marks

In another experiment, using the same fair tetrahedral die, we roll the die **two times** and define the following two events:

- **Event A** – The sum for the two rolls is odd.

- **Event B** – The sum for the two rolls is ≥ 6 .

- (b) Calculate the probability $P(A \cap B)$. 2 Marks

- (c) Calculate the probability $P(A \cup B)$. 3 Marks

- (d) Show mathematically whether Event A and Event B are independent or not. Justify your answer. 3 Marks

Question 2 [15 points]

Conditional Probability and Bayes Rule

A Habib University student commutes to the university (from her home) using one of the following options:

- **Option 1:** Takes the home car to the university 50% of the time. Under this option, she has an 85% chance of reaching the university in time and her commute time can be modelled as a uniformly distributed discrete random variable between 15 to 25 minutes.
- **Option 2:** A friend gives her a ride 25% of the time. Under this option, she has a 50% chance of reaching the university in time and her commute time can be modelled as a uniformly distributed discrete random variable between 10 to 15 minutes.
- **Option 3:** She takes a ride share option 25% of the time. Under this option, she has a 50% chance of reaching the university in time and her commute time can be modelled as a uniformly distributed discrete random variable between 20 to 25 minutes.

- (a) Calculate the probability that the student on a given day reaches the university late. 5 Marks

- (b) Given that she reached the university in time on a particular day, what is the probability that her friend gave her a ride to the campus that day? 5 Marks

- (c) Using the Total Expectation Theorem, calculate the expected commute time it takes for her to reach the campus. 5 Marks

Question 3 [15 Points]

Counting Problem

Habib University's innovation fair allows students to participate in three different activities. Each activity has its own independent selection process.

Activity A: Workshop Selection

There are 8 different technical workshops offered. A student must register for exactly 3 workshops, but:

- Workshop W1 and W2 cannot both be chosen (mutually exclusive).
- Workshop W3 is mandatory.

(a) How many valid workshop selections are possible? 5 Marks

Activity B: Presentation Slot

A student's presentation identity consists of:

- 1 title from 5 options
- 1 theme color from 4 options
- 1 format from 3 formats

However:

- If the student chooses Format 1, then the theme color cannot be Color 4.
- If the student chooses Title 5, then the theme color must be either Color 1 or Color 2.

(b) How many distinct valid presentations can be created? 5 Marks

Activity C: Access Code Creation

A 5-character access code is formed using digits (0–9) and uppercase letters (A–Z) with repetition allowed.

Rules:

- The code must contain at least one digit.
- The code must contain exactly one vowel among: A, E, I, O, U.

(c) How many valid access codes are possible? 5 Marks

Question 4 [20 points]

Distribution & Conditional Distribution with Discrete Random Variables

An HU student was intrigued in creating a 3-sided biased die (with numbers 1, 2, 3) and came up with an interesting idea. He developed a fair 6-sided die (each of the 6 sides with equal probability) and instead of numbers 1 to 6 appearing on each side, he engraved Number 1 on one of the 6 sides, Number 2 on two of the 6 sides and Number 3 on the three remaining sides. He rightly argued that this was a 3-sided biased die.

You roll this 3-sided biased die two times, noting the value of the **DRV (discrete random variable)** X (X is the number of times the Number 3 appears in the two rolls of the biased 3-sided die).

- (a) Calculate and plot the P.M.F., $P_X(X)$ for the DRV (discrete random variable) X .

4 Marks

(Hint – X can take on values 0, 1, and 2.)

We now define a new DRV Y , where the conditional PMF $P(Y|X = x)$ is a discrete uniform distribution taking on values x to 3. (Therefore, if R.V $X = 0$, then the conditioned R.V $Y|X = 0$ has a discrete uniform distribution taking on values 0, 1, 2, 3, each with probability 0.25. Similarly, if the R.V $X = 2$, then the conditioned R.V $Y|X = 2$ has a discrete uniform distribution taking on values 2, 3.)

- (b) Calculate and plot the conditional P.M.F. $P_{Y|X}(y|x)$ for the D.V Y . 8 Marks

(Hint – There will be three conditional PMFs $P_{Y|X}(y|x)$ corresponding to $X = 0, 1$, and 2.)

- (c) Calculate and plot the P.M.F. $P_Y(y)$ for the random variable Y . 8 Marks

(Hint – Use the above answers and the Total Probability Theorem to calculate $P_Y(y)$ for all values of y , where $y = 0, 1, 2, 3$.)